



Md. Rafej Khan

Python Developer | AI/ML Engineer | Deep Learning Researcher

PROFESSIONAL SUMMARY

Results-driven Python Developer and AI/ML Engineer with hands-on experience building production-ready machine learning systems, deep learning architectures, and intelligent automation tools. Proficient across the full ML pipeline — from data preprocessing and feature engineering to model deployment on AWS and Google Cloud. Published researcher in Quantum Neural Networks and contributor to open-source AI projects. Passionate about solving real-world problems with clean, scalable Python code.

WORK EXPERIENCE

Independent Python & AI/ML Developer

Self-Employed / Freelance • 2022 – Present • Full Time

- Architected and deployed end-to-end ML pipelines (data ingestion, preprocessing, model training, deployment) for real-world applications, hosted on AWS and Google Cloud.
- Engineered deep learning models using TensorFlow and PyTorch — CNNs, RNNs — achieving strong performance on classification and NLP tasks, NLP, and forecasting tasks.
- Built and fine-tuned LLM-based chatbots using OpenAI API and GPT architectures with RAG patterns for domain-specific knowledge retrieval.
- Conducted published research comparing Quantum Neural Networks vs. Classical ML models across multiple benchmarks including MNIST, healthcare datasets, and TSP.
- Contributed to the open-source AI community via GitHub repositories, research blogs, and documented Python packages.

KEY PROJECTS

JARVIS-Inspired AI Virtual Assistant

Python, OpenAI API, TensorFlow, NLP, Speech Recognition

- Built a voice-activated AI assistant capable of natural language understanding, web search, task automation, and real-time API integrations.
- Integrated speech-to-text and TTS modules; leveraged OpenAI API for contextual conversation management.
- Modular Python architecture with plugin-based command system — extensible to new skills without code refactoring.

Stock Price Prediction with Sentiment Analysis

Python, TensorFlow, RNN/LSTM, NLP, REST APIs

- Designed a multi-input LSTM model combining historical OHLCV data with NLP-extracted sentiment scores from real-time financial news.
- Improved prediction accuracy by 15% over baseline models by fusing price signals with news sentiment embeddings.
- Deployed on AWS Lambda with scheduled data ingestion from financial APIs.

Disease Detection & Medicine Recommendation System

Python, scikit-learn, TensorFlow, Pandas

- Built two companion ML models: a disease predictor (SVM + Decision Trees + Deep Learning ensemble) and a symptom-based medicine recommender.
- Trained on real-world healthcare datasets; packaged as a clean Python REST API consumable by frontend or mobile clients.

LLM-Powered Customer Support Chatbot

Python, OpenAI API, GPT, NLP, RNN

- Developed a production-grade chatbot using GPT-based models with custom prompt engineering for domain-specific support workflows.
- Benchmarked against RNN baseline — GPT integration improved contextual response quality by 40%+ on human evaluation.

Movie Recommendation Engine

Python, Django, scikit-learn, HTML/CSS/JS, Bootstrap-5

- Built a full-stack recommendation system with Django backend and responsive Bootstrap frontend.

CONTACT

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CORE SKILLS

Programming Languages

Python, C++, C, SQL, Bash

AI / ML Frameworks

TensorFlow, PyTorch, scikit-learn, Keras, OpenCV, CNNs, RNNs

NLP & LLMs

NLTK, spaCy, Prompt Engineering

Data Science

Pandas, NumPy, Feature Engineering

Cloud & DevOps

AWS, Google Cloud, Git, GitHub

Web & APIs

Django, REST APIs, HTML/CSS

Databases

PostgreSQL, MySQL

EDUCATION

B.Sc. in CSE

Eastern University, Dhaka

HSC — Science

South Asian College, Chittagong

CERTIFICATIONS

Software Testing & QA

ICT Division, Bangladesh Computer Council & EDGE

Machine Learning

Udemy

LANGUAGES

English — Fluent
Bengali — Native

- Implemented content-based filtering with cosine similarity; system handles 10,000+ movie records with sub-second response.

RESEARCH & PUBLICATIONS

Thesis: “Advancing Complex Problem Solving in AI: A Study of Quantum Neural Networks vs. Classical Models”

- Benchmarked QNNs against CNNs, RNNs, and hybrid models on MNIST, Breast Cancer, GPT text generation, MQ9 Weather, and TSP datasets.
- Evaluated models across accuracy, computational efficiency, scalability, and energy consumption — demonstrating hybrid models as optimal for near-term quantum advantage.